

# Chapter 6: Orthogonal Polynomials

## Mathematical Physics

Compiled by Jake W. Liu

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# Outline

- 1 From Laplace to ordinary Legendre
- 2 Legendre polynomials
- 3 Associated Legendre and spherical harmonics
- 4 Hermite polynomials
- 5 Laguerre polynomials
- 6 Chebyshev polynomials
- 7 Summary

# Where we are going

This chapter studies four families of **orthogonal polynomials** that arise by separating variables in Laplace, Poisson, Helmholtz, and Schrödinger problems. “Orthogonal” means a weighted inner product of distinct-degree polynomials vanishes; the interval and weight depend on the family.

- **Legendre**  $P_n$ . From Laplace in spherical coordinates to the ordinary ( $m = 0$ ) Legendre equation, series termination, generating function, recurrence, Rodrigues, orthogonality, and Fourier–Legendre expansions. Associated Legendre functions and spherical harmonics  $Y_n^m$  for  $m \neq 0$ , with the degree  $n$  the same integer that the axisymmetric problem picks out through  $\lambda = n(n + 1)$ .
- **Hermite**  $H_n$ . The Hermite equation and series solution, quantum harmonic oscillator motivation, generating function, Rodrigues formula, and weighted orthogonality.
- **Laguerre**  $L_n$  and **Chebyshev**  $T_n$ . Defining ODEs, first polynomials, weights and orthogonality, recurrences, and useful formula sheets.

A comparison table at the end collects interval, weight, and ODE for each family.

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# Two-dimensional Laplace equation in polar coordinates

**Step 1: separate the angular and radial parts.** A function satisfying Laplace's equation  $\nabla^2\psi = 0$  is called a *harmonic function*. The operator  $\nabla^2$  is the Laplacian:  $\psi_{xx} + \psi_{yy}$  in Cartesian coordinates. In polar coordinates  $r$  is radius and  $\theta$  is angle. We use its coordinate form without deriving it:

$$\nabla^2\psi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial\psi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2\psi}{\partial\theta^2} = 0. \quad (6.1.1)$$

Take a separated solution  $\psi(r, \theta) = u(r)v(\theta)$ . Then

$$\frac{\partial\psi}{\partial r} = u'(r)v(\theta), \quad \frac{\partial^2\psi}{\partial\theta^2} = u(r)v''(\theta),$$

so substitution into (6.1.1) gives

$$\frac{v}{r} \frac{d}{dr} (ru') + \frac{u}{r^2} v'' = 0.$$

Multiplying by  $r^2/(uv)$  yields

$$\frac{r(ru')'}{u} + \frac{v''}{v} = 0,$$

# Two-dimensional Laplace equation in polar coordinates (cont.)

and therefore

$$-\frac{v''}{v} = \frac{r(ru')'}{u} = \lambda, \quad (6.1.2)$$

because the left member depends only on  $\theta$  and the right member only on  $r$ .

The angular equation is

$$v'' + \lambda v = 0.$$

Periodicity  $v(\theta + 2\pi) = v(\theta)$  forces  $\lambda = n^2$  with  $n = 0, 1, 2, \dots$ : for  $\lambda = n^2 > 0$  the solutions  $A \cos(n\theta) + B \sin(n\theta)$  are  $2\pi$ -periodic; for  $\lambda = 0$ ,  $v = A + B\theta$  reduces to the constant  $n = 0$  mode; and  $\lambda < 0$  gives nonperiodic exponentials. The radial equation

$$r(ru')' - n^2 u = 0$$

is Cauchy–Euler; expanding the derivative,

$$r^2 u'' + ru' - n^2 u = 0, \quad (6.1.3)$$

and trying  $u = r^p$  gives  $(p^2 - n^2)r^p = 0$ , so  $p = \pm n$  for  $n \geq 1$ . For  $n = 0$ ,  $(ru')' = 0$  integrates to  $u(r) = D_0 + C_0 \log r$ .

## Two-dimensional Laplace equation in polar coordinates (cont.)

On an annulus, all these radial solutions are allowed. Inside a disk we must also examine the center. Inside a disk containing  $r = 0$ , boundedness removes  $\log r$  and all  $r^{-n}$  terms:

$$\psi(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^n [A_n \cos(n\theta) + C_n \sin(n\theta)]. \quad (6.1.4)$$

Equivalently, with  $z = re^{i\theta}$ ,  $c_0 = A_0$ , and  $c_n = A_n - iC_n$  for  $n \geq 1$ ,

$$\psi(r, \theta) = \operatorname{Re} \left( \sum_{n=0}^{\infty} c_n z^n \right), \quad z^n = r^n e^{in\theta}.$$

Thus the regular real harmonic functions are real parts of analytic power series; exterior or annular domains additionally allow negative powers and, for the angularly independent  $n = 0$  mode, a logarithm.

# Three-dimensional Laplace equation in spherical coordinates

In spherical coordinates  $r$  is radius,  $\theta$  is the polar angle measured from the positive vertical axis, and  $\phi$  is the azimuthal angle around that axis. We use the spherical Laplacian without deriving this coordinate formula:

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} = 0. \quad (6.1.5)$$

**Step 1: separate the radial variable.** Set  $\psi(r, \theta, \phi) = u(r)y(\theta, \phi)$ . Because  $y$  is independent of  $r$  and  $u$  is independent of the angles,

$$\begin{aligned} \frac{\partial}{\partial r} (r^2 \psi_r) &= y(r^2 u')', \\ \frac{\partial}{\partial \theta} (\sin \theta \psi_\theta) &= u(\sin \theta y_\theta)_\theta. \end{aligned}$$

Substitution into (6.1.5) and multiplication by  $r^2/(uy)$  give

$$\frac{(r^2 u')'}{u} = - \frac{\frac{1}{\sin \theta} (\sin \theta y_\theta)_\theta + \frac{1}{\sin^2 \theta} y_{\phi\phi}}{y} = \lambda. \quad (6.1.6)$$

# Three-dimensional Laplace equation in spherical coordinates (cont.)

The first expression depends only on  $r$  and the second only on  $(\theta, \phi)$ , so each equals a constant  $\lambda$ .

**Step 2: radial Euler equation.** Expanding  $(r^2 u')' = r^2 u'' + 2ru'$  gives

$$r^2 u'' + 2ru' - \lambda u = 0.$$

Try  $u = r^\sigma$ . Then  $u' = \sigma r^{\sigma-1}$ ,  $u'' = \sigma(\sigma-1)r^{\sigma-2}$ , and

$$[\sigma(\sigma+1) - \lambda] r^\sigma = 0.$$

So  $\lambda = \sigma(\sigma+1)$ . Once the angular problem forces  $\lambda = n(n+1)$  with  $n = 0, 1, 2, \dots$  (next frames), the two exponents are

$$\sigma = n, \quad \sigma = -n-1,$$

and

$$u_n(r) = A_n r^n + B_n r^{-n-1}. \tag{6.1.7}$$

The  $r^n$  branch is regular at the origin; the  $r^{-n-1}$  branch decays at infinity.

# Three-dimensional Laplace equation in spherical coordinates (cont.)

The remaining angular eigenproblem is

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 y}{\partial \phi^2} + \lambda y = 0.$$

We first treat the **axisymmetric** case (no  $\phi$ -dependence), which yields the ordinary Legendre equation and  $\lambda = n(n+1)$ . Associated Legendre modes for  $m \neq 0$  come later as a short extension.

# Axisymmetry: the ordinary Legendre equation

**Step 1: assume no azimuthal dependence and change variables.** If  $\psi$  is independent of  $\phi$  (axisymmetry about the polar axis), the angular equation reduces to

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left( \sin \theta \frac{dv}{d\theta} \right) + \lambda v = 0.$$

Set  $x = \cos \theta$ , so  $dx/d\theta = -\sin \theta$  and  $d/d\theta = -\sin \theta d/dx$ . Then

$$\begin{aligned} \sin \theta \frac{dv}{d\theta} &= -(1-x^2)v_x, \\ \frac{d}{d\theta} \left( \sin \theta \frac{dv}{d\theta} \right) &= \sin \theta \frac{d}{dx} [(1-x^2)v_x], \end{aligned}$$

and therefore

$$\frac{d}{dx} [(1-x^2)v'(x)] + \lambda v(x) = 0. \quad (6.1.8)$$

Expanding the derivative gives the equivalent form

$$(1-x^2)v'' - 2xv' + \lambda v = 0. \quad (6.1.9)$$

(The poles  $\theta = 0, \pi$  correspond to the endpoints  $x = \pm 1$ .)

# Axisymmetry: the ordinary Legendre equation (cont.)

**Step 2: power series and coefficient recurrence.** Set

$$v(x) = \sum_{j=0}^{\infty} a_j x^j.$$

Then

$$v' = \sum_{j=0}^{\infty} (j+1) a_{j+1} x^j,$$

$$v'' = \sum_{j=0}^{\infty} (j+2)(j+1) a_{j+2} x^j,$$

$$-x^2 v'' = - \sum_{j=0}^{\infty} j(j-1) a_j x^j,$$

$$-2xv' = -2 \sum_{j=0}^{\infty} j a_j x^j.$$

## Axisymmetry: the ordinary Legendre equation (cont.)

Substitution into (6.1.9) gives

$$\begin{aligned} 0 &= \sum_{j=0}^{\infty} [(j+2)(j+1)a_{j+2} - j(j-1)a_j - 2ja_j + \lambda a_j] x^j \\ &= \sum_{j=0}^{\infty} [(j+2)(j+1)a_{j+2} + (\lambda - j(j+1))a_j] x^j. \end{aligned}$$

Hence the coefficient recurrence is

$$a_{j+2} = -\frac{\lambda - j(j+1)}{(j+2)(j+1)} a_j. \quad (6.1.10)$$

Even and odd chains are independent.

**Step 3: select solutions that remain finite at the poles.** The points  $x = \pm 1$  correspond to the north and south poles of the sphere. A polynomial is finite at both. The recurrence stops at degree  $n$  precisely when  $\lambda = n(n+1)$ ; this is the simplest way to construct the acceptable solutions.

We use the standard endpoint result without its proof: a nonzero solution finite at both endpoints exists only for these values of  $\lambda$ , and is a multiple of that polynomial. The following example explains why convergence only inside  $(-1, 1)$  is not enough.

## Axisymmetry: the ordinary Legendre equation (cont.)

*Example.* Take  $\lambda = 2$  and start the *even* chain at  $a_0 = 1$ . Then (6.1.10) gives

$$a_2 = -\frac{2-0}{2 \cdot 1} = -1, \quad a_4 = -\frac{2-6}{4 \cdot 3}(-1) = -\frac{1}{3}, \quad a_6 = -\frac{2-20}{6 \cdot 5} \left(-\frac{1}{3}\right) = -\frac{1}{5},$$

and the pattern is  $a_{2k} = -1/(2k-1)$  for  $k \geq 1$ . Subtracting the Taylor series for  $\log(1+x)$  and  $\log(1-x)$  gives  $\sum_{k \geq 1} x^{2k-1}/(2k-1) = \frac{1}{2} \log[(1+x)/(1-x)]$  for  $|x| < 1$ . Hence

$$v(x) = 1 - \sum_{k \geq 1} \frac{x^{2k}}{2k-1} = 1 - x \sum_{k \geq 1} \frac{x^{2k-1}}{2k-1} = 1 - \frac{x}{2} \log \frac{1+x}{1-x}.$$

This is a genuine solution of (6.1.9) with  $\lambda = 2$  (a one-line check by substitution), finite at every interior point, yet  $v(x) \rightarrow -\infty$  as  $x \rightarrow 1^-$ : useless on the sphere. Meanwhile the *odd* chain with  $\lambda = 2$  terminates immediately and gives the polynomial  $v = x$ . For the allowed value  $\lambda = n(n+1)$ , the recurrence factors as

$$a_{j+2} = -\frac{(n-j)(n+j+1)}{(j+2)(j+1)} a_j. \quad (6.1.11)$$

## Axisymmetry: the ordinary Legendre equation (cont.)

At  $j = n$  one has  $a_{n+2} = 0$ , so the same-parity chain is a polynomial of degree  $n$ . Set the initial coefficient of the opposite-parity chain to zero. The surviving polynomial is bounded at both poles.

With  $\lambda = n(n+1)$ , equations (6.1.8)–(6.1.9) become the ordinary Legendre equation

$$\frac{d}{dx} [(1-x^2)v'(x)] + n(n+1)v(x) = 0, \quad (6.1.12)$$

or

$$(1-x^2)v'' - 2xv' + n(n+1)v = 0. \quad (6.1.13)$$

Choose the scale so that  $P_n(1) = 1$ ; the resulting polynomial is the *Legendre polynomial*  $P_n$ . Since the recurrence advances by two and only the terminating parity chain is retained, its powers are  $x^n, x^{n-2}, \dots$ . Thus

$$P_n(-x) = (-1)^n P_n(x).$$

Rodrigues' formula below supplies the normalization directly. The generating function below fixes all the normalized coefficients explicitly.

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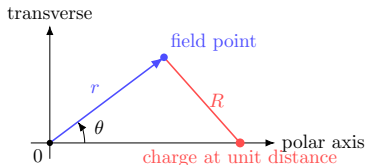
# Generating function from the Coulomb potential

**Step 1: write the axisymmetric potential.** Place a point charge one unit from the origin on the positive polar axis. For a field point with spherical coordinates  $(r, \theta, \phi)$ , the law of cosines gives

$$R^2 = r^2 + 1^2 - 2r(1) \cos \theta = 1 - 2r \cos \theta + r^2,$$

so, for a nonzero physical scale factor  $C$ , the Coulomb potential is

$$\varphi(r, \theta) = \frac{C}{R} = \frac{C}{\sqrt{1 - 2r \cos \theta + r^2}}. \quad (6.2.1)$$



$$R^2 = 1 + r^2 - 2r \cos \theta$$

# Generating function from the Coulomb potential (cont.)

In the picture the horizontal line is the polar axis  $\theta = 0$ , along which the charge sits, and  $\theta$  is measured from it exactly as in (6.1.5). That axis is a genuine Cartesian direction; do not confuse it with the variable  $x = \cos \theta$  used in the formulas of this chapter, which is the cosine of that angle and ranges over  $[-1, 1]$ .

For  $r < 1$  the source is outside the region under study. The potential is harmonic, finite at the origin, and independent of  $\phi$ . Our separated solutions therefore suggest the expansion

$$\frac{1}{\sqrt{1 - 2r \cos \theta + r^2}} = \sum_{n=0}^{\infty} a_n r^n P_n(\cos \theta). \quad (6.2.2)$$

This is a physical motivation for the generating function. We quote the existence of this convergent expansion for  $r < 1$ ; the axis calculation now fixes its coefficients.

**Step 2: determine the coefficients on the axis.** Set  $\theta = 0$ . Then  $\cos \theta = 1$ ,  $P_n(1) = 1$ , and

$$1 - 2r \cos 0 + r^2 = (1 - r)^2.$$

Because  $0 \leq r < 1$ ,  $|1 - r| = 1 - r$ , so (6.2.2) becomes

$$\frac{1}{1 - r} = \sum_{n=0}^{\infty} a_n r^n = \sum_{n=0}^{\infty} r^n \quad (|r| < 1).$$

# Generating function from the Coulomb potential (cont.)

Uniqueness of power-series coefficients gives  $a_n = 1$ . Writing  $t = r$  and  $x = \cos \theta$ ,

$$g(x, t) = \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n, \quad |t| < 1, \quad -1 \leq x \leq 1. \quad (6.2.3)$$

**Step 3: expand the first three  $P_n$  from the generating function.** For small  $t$ , write

$$g(x, t) = (1 - u)^{-1/2}, \quad u = 2xt - t^2,$$

and use the binomial series

$$(1 - u)^{-1/2} = 1 + \frac{1}{2}u + \frac{1}{2} \cdot \frac{3}{2} \frac{u^2}{2!} + O(u^3) = 1 + \frac{1}{2}u + \frac{3}{8}u^2 + O(u^3).$$

Collect powers of  $t$  up to  $t^2$ :

$$\begin{aligned} u &= 2xt - t^2, \\ u^2 &= (2xt - t^2)^2 = 4x^2t^2 + O(t^3). \end{aligned}$$

# Generating function from the Coulomb potential (cont.)

Hence

$$\begin{aligned}g &= 1 + \frac{1}{2}(2xt - t^2) + \frac{3}{8}(4x^2t^2) + O(t^3) \\&= 1 + xt + \left(-\frac{1}{2} + \frac{3}{2}x^2\right)t^2 + O(t^3) \\&= 1 + xt + \frac{1}{2}(3x^2 - 1)t^2 + O(t^3).\end{aligned}$$

Matching  $\sum P_n(x)t^n$  therefore gives the first three polynomials explicitly:

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1). \quad (6.2.4)$$

Checks:  $P_n(1) = 1$  and  $P_n(-x) = (-1)^n P_n(x)$  hold for  $n = 0, 1, 2$ .

# Three-term recurrence

**Step 1: differentiate the generating function.** Let

$$D(x, t) = 1 - 2xt + t^2, \quad g(x, t) = D(x, t)^{-1/2}.$$

Differentiating with respect to  $t$  gives

$$\frac{\partial g}{\partial t} = -\frac{1}{2}D^{-3/2}\frac{\partial D}{\partial t} = \frac{x-t}{D^{3/2}}.$$

Termwise differentiation of the power series gives

$$\frac{\partial g}{\partial t} = \sum_{n=1}^{\infty} nP_n(x)t^{n-1}. \quad (6.2.5)$$

Multiplying  $g_t = (x-t)D^{-3/2}$  by  $D$  gives  $(x-t)g = Dg_t$ . Substituting both series,

$$(x-t)\sum_{n=0}^{\infty} P_nt^n = (1-2xt+t^2)\sum_{n=1}^{\infty} nP_nt^{n-1},$$

where  $P_n = P_n(x)$ . Expanding,

$$\sum_{n=0}^{\infty} xP_nt^n - \sum_{n=0}^{\infty} P_nt^{n+1} = \sum_{n=1}^{\infty} nP_nt^{n-1} - 2\sum_{n=1}^{\infty} n xP_nt^n + \sum_{n=1}^{\infty} nP_nt^{n+1}. \quad (6.2.6)$$

## Three-term recurrence (cont.)

**Step 2: compare coefficients of  $t^n$ .** With the convention  $P_{-1} = 0$ , the left coefficient of  $t^n$  is  $xP_n - P_{n-1}$ . On the right,

$$[t^n] \sum_{j \geq 1} j P_j t^{j-1} = (n+1)P_{n+1}, \quad [t^n] \left( -2 \sum_{j \geq 1} j x P_j t^j \right) = -2n x P_n,$$

and

$$[t^n] \sum_{j \geq 1} j P_j t^{j+1} = (n-1)P_{n-1}$$

(using  $P_{-1} = 0$ ). Equating coefficients,

$$xP_n - P_{n-1} = (n+1)P_{n+1} - 2n x P_n + (n-1)P_{n-1}.$$

Collecting the  $P_n$  and  $P_{n-1}$  terms gives

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x), \quad n \geq 0, \quad (6.2.7)$$

with  $P_{-1} = 0$  and  $P_0 = 1$ . For  $n = 0$ ,  $P_1 = xP_0 = x$ .

# First few Legendre polynomials

We already obtained  $P_0 = 1$ ,  $P_1 = x$ , and  $P_2 = (3x^2 - 1)/2$  from the generating function. The recurrence (6.2.7) supplies the next one without another series expansion:

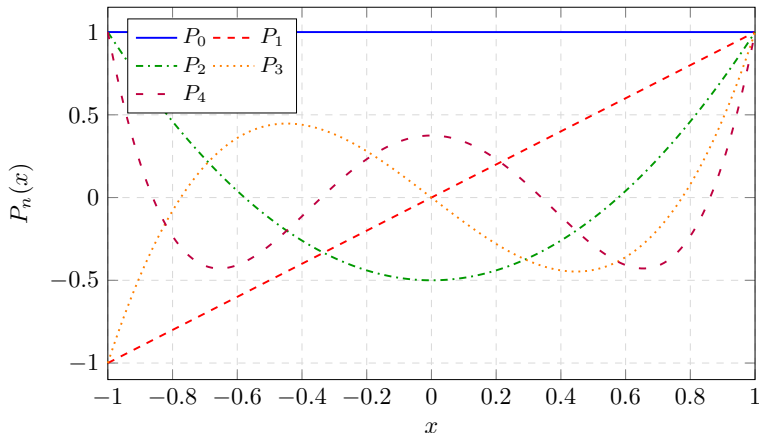
$$3P_3 = 5xP_2 - 2P_1 = \frac{5x(3x^2 - 1)}{2} - 2x = \frac{15x^3 - 9x}{2}.$$

Thus

$$P_3(x) = \frac{1}{2}(5x^3 - 3x).$$

It has odd parity and  $P_3(1) = 1$ , as expected. The following plot shows how the oscillations change with degree.

# Shapes of the first Legendre polynomials



Each  $P_n$  is bounded on  $[-1, 1]$  and normalized by  $P_n(1) = 1$ . Increasing  $n$  adds interior zeros while preserving  $\int_{-1}^1 P_m(x)P_n(x) dx = 0$  for  $m \neq n$ .

# Rodrigues' formula

The recurrence generates  $P_0, P_1, P_2, \dots$  in order. A complementary formula constructs any one polynomial directly:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n. \quad (6.2.8)$$

This is Rodrigues' formula. To prove it, start with the corrected binomial expansion

$$(x^2 - 1)^n = \sum_{k=0}^n (-1)^k \binom{n}{k} x^{2(n-k)}.$$

After  $n$  derivatives, a term survives only when  $2(n-k) \geq n$ , or  $k \leq \lfloor n/2 \rfloor$ . Hence

$$\frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (2n-2k)!}{2^n k! (n-k)! (n-2k)!} x^{n-2k}.$$

To check the match with the power-series solution, call the coefficient of  $x^{n-2k}$  in this sum  $c_k$ . Consecutive coefficients satisfy

$$\frac{c_{k+1}}{c_k} = -\frac{(n-2k)(n-2k-1)}{(2k+2)(2n-2k-1)}.$$

## Rodrigues' formula (cont.)

This is (6.1.11) solved downward with  $j = n - 2k - 2$ . Thus the derivative expression has the same degree, parity, and coefficient recurrence as  $P_n$ . The normalization at  $x = 1$ , checked below, fixes the remaining common factor. The sign is easiest to remember in the form  $(x^2 - 1)^n$ ; equivalently,

$$P_n(x) = \frac{(-1)^n}{2^n n!} \frac{d^n}{dx^n} (1 - x^2)^n.$$

For  $n = 2$  either form returns  $P_2 = (3x^2 - 1)/2$ .

The factor  $2^n n!$  also verifies the normalization directly. In  $(x - 1)^n (x + 1)^n$ , the only term surviving after  $n$  derivatives at  $x = 1$  differentiates  $(x - 1)^n$  all  $n$  times and equals  $n! 2^n$ .

# Rodrigues' formula (cont.)

**Two consequences needed for the norm.** The highest power in  $(x^2 - 1)^n$  is  $x^{2n}$ . Hence

$$\frac{d^n}{dx^n} x^{2n} = \frac{(2n)!}{n!} x^n,$$

and division by  $2^n n!$  gives

$$P_n(x) = \frac{(2n)!}{2^n (n!)^2} x^n + \text{lower powers.} \quad (6.2.9)$$

Differentiating  $n$  more times removes all lower powers:

$$P_n^{(n)}(x) = \frac{(2n)!}{2^n n!}. \quad (6.2.10)$$

The zeros of order  $n$  in  $(x^2 - 1)^n$  at  $x = \pm 1$  also make the boundary terms vanish when we integrate by parts up to  $n$  times. These are the two facts used next.

**Connection to Chapter 1.** Cauchy's derivative formula applied to  $(t^2 - 1)^n$  also writes Rodrigues' formula as

$$P_n(x) = \frac{1}{2^{n+1}\pi i} \oint_C \frac{(t^2 - 1)^n}{(t - x)^{n+1}} dt, \quad (6.2.11)$$

where  $C$  is a simple counterclockwise contour enclosing  $x$ . We will use the derivative form for the norm calculation.

# Orthogonality and norm of Legendre polynomials

The Legendre equation is

$$\frac{d}{dx} [(1-x^2)P_n'(x)] + n(n+1)P_n(x) = 0. \quad (6.2.12)$$

This is the Chapter 4 problem with  $p = 1 - x^2$ ,  $q = 0$ , and  $w = 1$ . At either endpoint,

$$(1-x^2)(P_n P_m' - P_n' P_m) \longrightarrow 0,$$

since polynomials and their derivatives remain finite. The eigenvalues  $n(n+1)$  differ for distinct nonnegative degrees, so Chapter 4's Sturm–Liouville theorem already predicts orthogonality. A direct proof from Rodrigues' formula also supplies the norm.

Assume without loss of generality that  $m \geq n$ . Use Rodrigues' formula for the higher-degree polynomial:

$$\int_{-1}^1 P_n(x) P_m(x) dx = \frac{1}{2^m m!} \int_{-1}^1 P_n(x) \frac{d^m}{dx^m} (x^2 - 1)^m dx.$$

Because  $(x^2 - 1)^m = (x - 1)^m (x + 1)^m$  has a zero of order  $m$  at each endpoint,

$$\left. \frac{d^j}{dx^j} (x^2 - 1)^m \right|_{x=\pm 1} = 0, \quad 0 \leq j \leq m-1.$$

# Orthogonality and norm of Legendre polynomials (cont.)

All boundary terms therefore vanish in  $m$  successive integrations by parts:

$$\int_{-1}^1 P_n P_m dx = \frac{(-1)^m}{2^m m!} \int_{-1}^1 P_n^{(m)}(x) (x^2 - 1)^m dx. \quad (6.2.13)$$

If  $m > n$ , then  $P_n^{(m)} = 0$  because  $P_n$  has degree  $n$ . Thus

$$\int_{-1}^1 P_n(x) P_m(x) dx = 0 \quad (n \neq m). \quad (6.2.14)$$

# Orthogonality and norm of Legendre polynomials (cont.)

For the diagonal case set  $m = n$  in (6.2.13). Equation (6.2.10) gives  $P_n^{(n)} = (2n)!/(2^n n!)$ , while  $(-1)^n(x^2 - 1)^n = (1 - x^2)^n$ . Therefore

$$I_n := \int_{-1}^1 P_n(x)^2 dx = \frac{(2n)!}{2^{2n}(n!)^2} \int_{-1}^1 (1 - x^2)^n dx. \quad (6.2.15)$$

For  $n = 0$ ,  $I_0 = 2$  directly because  $P_0 = 1$ , so the calculation includes the constant polynomial as well.

# Orthogonality and norm of Legendre polynomials (cont.)

To evaluate the remaining integral, set  $x = 2u - 1$ . Then  $dx = 2 du$  and  $1 - x^2 = 4u(1 - u)$ , so with the beta function

$$B(a, b) = \int_0^1 u^{a-1}(1-u)^{b-1} du = \Gamma(a)\Gamma(b)/\Gamma(a+b),$$

$$\begin{aligned}\int_{-1}^1 (1-x^2)^n dx &= 2 \cdot 4^n \int_0^1 u^n (1-u)^n du = 2^{2n+1} B(n+1, n+1) \\ &= 2^{2n+1} \frac{(n!)^2}{(2n+1)!}.\end{aligned}$$

Therefore

$$I_n = \frac{(2n)!}{2^{2n}(n!)^2} \cdot \frac{2^{2n+1}(n!)^2}{(2n+1)!} = \frac{2}{2n+1}.$$

Combining orthogonality and norm,

$$\int_{-1}^1 P_n(x) P_m(x) dx = \frac{2}{2n+1} \delta_{nm}. \quad (6.2.16)$$

# Fourier–Legendre series

We use the standard completeness result without proof: Legendre polynomials form a complete orthogonal system in  $L^2([-1, 1])$ , just as the sines did in Chapter 4. With projection coefficients, their partial sums converge in the  $L^2$  norm. Thus, for  $f \in L^2([-1, 1])$ ,

$$f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$$

in the  $L^2$  sense. Multiply by  $P_m(x)$  and integrate:

$$\int_{-1}^1 f(x) P_m(x) dx = \sum_{n=0}^{\infty} a_n \int_{-1}^1 P_n(x) P_m(x) dx = a_m \frac{2}{2m+1}.$$

Solving for  $a_m$  gives

$$a_m = \frac{2m+1}{2} \int_{-1}^1 f(x) P_m(x) dx. \quad (6.2.17)$$

Therefore the Fourier–Legendre expansion is

$$f(x) = \sum_{n=0}^{\infty} \left[ \frac{2n+1}{2} \int_{-1}^1 f(s) P_n(s) ds \right] P_n(x). \quad (6.2.18)$$

# Useful formulas for the Legendre polynomial

differential equation  $(1 - x^2)P_n'' - 2x P_n' + n(n+1)P_n = 0$

series solution 
$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (2n-2k)!}{2^n k! (n-2k)! (n-k)!} x^{n-2k}$$

generating function 
$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n$$

recurrence  $(n+1)P_{n+1} = (2n+1)x P_n - n P_{n-1}$

orthogonality 
$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{nm}$$

Rodrigues 
$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Schl\"afli 
$$P_n(x) = \frac{1}{2^{n+1} \pi i} \oint \frac{(t^2 - 1)^n}{(t - x)^{n+1}} dt \quad (6.2.11)$$

The finite sum follows by expanding Rodrigues' formula;  $\lfloor n/2 \rfloor$  means the greatest integer no larger than  $n/2$ . Generating series:  $|t| < 1$  for  $x \in [-1, 1]$ . The contour encloses  $t = x$ .

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# Associated Legendre equation and $Y_n^m$

When the field depends on  $\phi$ , separate  $y(\theta, \phi) = v(\theta)w(\phi)$ . As in the polar calculation, the separated azimuthal modes can be chosen as  $w = e^{im\phi}$ ; periodicity  $w(\phi + 2\pi) = w(\phi)$  requires  $m \in \mathbb{Z}$ . Return to the angular equation in §6.1 and put  $y(\theta, \phi) = v(\theta)e^{im\phi}$ :

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 y}{\partial \phi^2} + \lambda y = 0.$$

Since  $\frac{\partial^2}{\partial \phi^2} e^{im\phi} = (im)^2 e^{im\phi} = -m^2 e^{im\phi}$ , every term carries the common factor  $e^{im\phi}$ , which is never zero and therefore cancels:

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left( \sin \theta \frac{dv}{d\theta} \right) + \left[ \lambda - \frac{m^2}{\sin^2 \theta} \right] v = 0.$$

Now set  $x = \cos \theta$ . The first term was already converted in the axisymmetric frame: there  $\sin \theta dv/d\theta = -(1 - x^2)v'$  gave

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left( \sin \theta \frac{dv}{d\theta} \right) = \frac{d}{dx} [(1 - x^2)v'(x)],$$

## Associated Legendre equation and $Y_n^m$ (cont.)

and for the new term  $\sin^2 \theta = 1 - \cos^2 \theta = 1 - x^2$ , so  $m^2 / \sin^2 \theta$  becomes  $m^2 / (1 - x^2)$ . The polar equation is therefore

$$\frac{d}{dx} [(1 - x^2)v'(x)] + \left[ \lambda - \frac{m^2}{1 - x^2} \right] v(x) = 0. \quad (6.3.1)$$

For a smooth angular field on the whole sphere, the allowed values are  $\lambda = n(n + 1)$  with  $n = |m|, |m| + 1, \dots$  (endpoint result quoted). The equation becomes

$$(1 - x^2)v'' - 2xv' + \left[ n(n + 1) - \frac{m^2}{1 - x^2} \right] v = 0. \quad (6.3.2)$$

Instead of repeating a long series analysis at the poles, construct these solutions from the Legendre polynomials already known.

# Associated Legendre equation and $Y_n^m$ (cont.)

**A derivative construction (quoted).** The associated Legendre functions on  $[-1, 1]$  are obtained from the known polynomials by

$$P_n^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad m = 0, 1, \dots, n. \quad (6.3.3)$$

The sign  $(-1)^m$  is a convention, called the Condon–Shortley phase. For example,  $P_1 = x$  and  $P_2 = (3x^2 - 1)/2$  give

$$P_1^1(x) = -\sqrt{1-x^2},$$

$$P_2^1(x) = -3x\sqrt{1-x^2},$$

$$P_2^2(x) = (1-x^2) \frac{d^2}{dx^2} \frac{3x^2-1}{2} = 3(1-x^2).$$

For  $m = 0$  this returns  $P_n$ . For  $m > n$ , the derivative is zero, so it produces no additional mode. This construction supplies the regular angular solutions selected above.

**Convention warning.** Much of the applied-scattering literature omits the Condon–Shortley factor, writing  $P_n^m(x) = (1-x^2)^{m/2} P_n^{(m)}(x)$ ; then  $P_n^1(\cos \theta) = \sin \theta P_n'(\cos \theta)$  (prime in the argument  $x = \cos \theta$ ), e.g.  $P_1^1(\cos \theta) = \sin \theta$ . Check the sign convention before combining formulas from different sources.

# Associated Legendre equation and $Y_n^m$ (cont.)

**Spherical harmonics: name the angular modes.** An unnormalized angular basis is

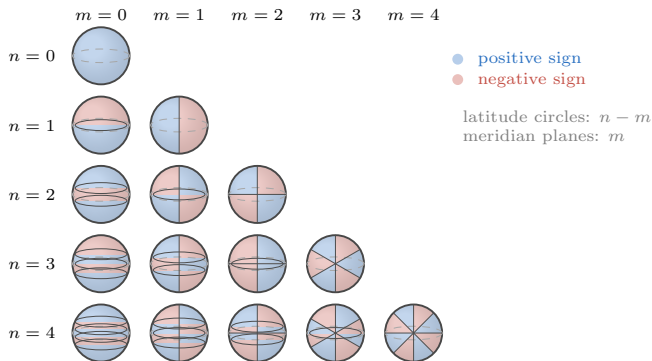
$$\mathcal{Y}_n^m(\theta, \phi) = P_n^{|m|}(\cos \theta) e^{im\phi}, \quad -n \leq m \leq n.$$

Normalized spherical harmonics  $Y_n^m$  differ from this by a constant and a fixed phase. Many texts call the degree  $\ell$  and write the harmonics as  $Y_{\ell m}$ ; here the degree is called  $n$ . With the same normalization and phase convention,  $Y_{\ell m}$  and  $Y_n^m$  denote the same function when  $\ell = n$ . Combining radial and angular factors, the regular interior expansion is

$$\psi_{\text{int}}(r, \theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm} r^n \mathcal{Y}_n^m(\theta, \phi), \quad (6.3.4)$$

and the decaying exterior expansion uses  $r^{-n-1}$  instead of  $r^n$ .

# Spherical-harmonic nodal patterns



Schematic counts, not exact nodal locations. Rows: degree  $n$ ; columns: order  $m$ . The factor  $P_n^{|m|}(\cos \theta)$  has  $n - |m|$  interior nodal latitudes. For  $m > 0$ , a real factor  $\cos(m\phi)$  or  $\sin(m\phi)$  adds  $m$  meridian planes; for  $m = 0$ , there are none.

# When the degree is not an integer

The Legendre equation makes sense with  $\lambda = \nu(\nu + 1)$  for noninteger degree  $\nu$ . Integer  $n$  was selected above by regularity at *both* endpoints  $x = \pm 1$ ; requiring regularity only at  $x = 1$  leaves the Legendre function of the first kind  $P_\nu$ , which for noninteger degree is logarithmically singular at  $x = -1$ . Wave problems use the reflected argument  $P_\nu(-\cos \theta)$ , which is regular at  $\theta = \pi$ .

# From scalar harmonics to tangential vector fields

Vector fields on a sphere are built from the scalar modes by curls; the angular factors remain the associated Legendre modes  $P_n^{|m|}(\cos \theta)e^{im\phi}$ , and the radial factors are the spherical Bessel and Hankel functions  $j_n$  and  $h_n^{(1,2)}$  of Chapter 5. No new special functions appear.

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# The Hermite equation and series solution

**Step 1: substitute a power series.** The physicists' Hermite polynomials solve

$$y'' - 2xy' + 2ny = 0, \quad n = 0, 1, 2, \dots \quad (6.4.1)$$

Let  $y(x) = \sum_{j=0}^{\infty} a_j x^j$ . Then

$$y' = \sum_{j=0}^{\infty} (j+1) a_{j+1} x^j,$$

$$y'' = \sum_{j=0}^{\infty} (j+2)(j+1) a_{j+2} x^j,$$

$$-2xy' = -2 \sum_{j=0}^{\infty} j a_j x^j.$$

Substitution into (6.4.1) gives

$$\sum_{j=0}^{\infty} [(j+2)(j+1)a_{j+2} + 2(n-j)a_j] x^j = 0.$$

# The Hermite equation and series solution (cont.)

Every coefficient vanishes, so

$$a_{j+2} = \frac{2(j-n)}{(j+2)(j+1)} a_j. \quad (6.4.2)$$

**Step 2: termination condition.** Even and odd chains are independent. For integer  $n \geq 0$ , set  $j = n$  in (6.4.2):

$$a_{n+2} = \frac{2(n-n)}{(n+2)(n+1)} a_n = 0.$$

Thus the same-parity chain terminates at degree  $n$ , giving a polynomial of exact degree  $n$ . The opposite-parity chain never hits a zero numerator, so we set its initial coefficient to zero (the non-polynomial second solution of (6.4.1)).

**Step 3: explicit series from the downward product.** Adopt the physicists' leading coefficient  $a_n = 2^n$ . The two-step recurrence (6.4.2) run *downward* from degree  $n$  is

$$a_j = \frac{(j+2)(j+1)}{2(j-n)} a_{j+2} \quad (j = n-2, n-4, \dots).$$

# The Hermite equation and series solution (cont.)

The first two downward steps are

$$a_{n-2} = -\frac{n(n-1)}{4} a_n, \quad a_{n-4} = -\frac{(n-2)(n-3)}{8} a_{n-2},$$

whenever the indicated indices are nonnegative. After  $k$  steps, the numerator contains the first  $2k$  descending factors of  $n!$ , and the denominator is  $(-4)(-8)\cdots(-4k) = (-4)^k k!$ . Hence

$$a_{n-2k} = \frac{2^n n!}{(-4)^k k! (n-2k)!} = \frac{(-1)^k 2^{n-2k} n!}{k! (n-2k)!}.$$

Writing  $a_{n-2k} x^{n-2k} = (-1)^k (2x)^{n-2k} n! / (k! (n-2k)!)$  and summing gives

$$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k (2x)^{n-2k} \frac{n!}{k! (n-2k)!}. \quad (6.4.3)$$

Low- $n$  checks:

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x.$$

# The Hermite equation and series solution (cont.)

Multiplying (6.4.1) by  $e^{-x^2}$  yields the Sturm–Liouville form

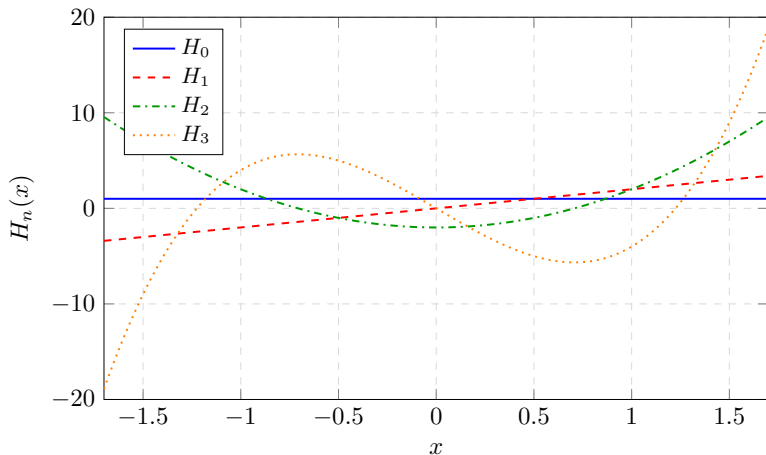
$$\frac{d}{dx} \left( e^{-x^2} y' \right) + 2n e^{-x^2} y = 0. \quad (6.4.4)$$

The same Lagrange-identity argument used for Legendre shows, for  $n \neq m$ ,

$$\int_{-\infty}^{\infty} H_n(x) H_m(x) e^{-x^2} dx = 0. \quad (6.4.5)$$

(The diagonal norm is computed from the generating function below.)

# Shapes of the first Hermite polynomials



Hermite polynomials grow away from the origin; the physical oscillator functions are  $e^{-x^2/2}H_n(x)$ .

# Quantum harmonic oscillator

For a mass- $M$  particle in  $V(x) = \frac{1}{2}M\omega^2x^2$ , the time-independent Schrödinger equation is given below. Here  $E$  is the energy,  $\omega > 0$  is the oscillator's angular frequency,  $\hbar$  is the reduced Planck constant, and  $\psi$  is the spatial wavefunction:

$$-\frac{\hbar^2}{2M}\psi'' + \frac{1}{2}M\omega^2x^2\psi = E\psi.$$

Use  $\xi = \sqrt{M\omega/\hbar}x$ . Then

$$\frac{d^2\psi}{dx^2} = \frac{M\omega}{\hbar} \frac{d^2\psi}{d\xi^2}, \quad \frac{1}{2}M\omega^2x^2 = \frac{1}{2}\hbar\omega\xi^2.$$

Substitution and division by  $\hbar\omega/2$  give  $-\psi_{\xi\xi} + \xi^2\psi = (2E/\hbar\omega)\psi$ , or

$$\frac{d^2\psi}{d\xi^2} + (\varepsilon - \xi^2)\psi = 0, \quad \varepsilon = \frac{2E}{\hbar\omega}. \quad (6.4.6)$$

For large  $|\xi|$ ,  $\psi'' - \xi^2\psi \approx 0$  suggests the decaying factor  $e^{-\xi^2/2}$ . Set

$$\psi(\xi) = e^{-\xi^2/2}y(\xi).$$

# Quantum harmonic oscillator (cont.)

Compute the derivatives by the product rule. Write  $f := e^{-\xi^2/2}$ , so  $f' = -\xi f$  and  $f'' = (-1 + \xi^2)f$  (because  $f'' = -(f + \xi f') = -(f + \xi(-\xi f)) = (-1 + \xi^2)f$ ). Then

$$\begin{aligned}\psi' &= f'y + fy' = e^{-\xi^2/2}(y' - \xi y), \\ \psi'' &= f''y + 2f'y' + fy'' \\ &= e^{-\xi^2/2} \left[ (-1 + \xi^2)y + 2(-\xi)y' + y'' \right] \\ &= e^{-\xi^2/2} (y'' - 2\xi y' + (\xi^2 - 1)y).\end{aligned}$$

Substitute into (6.4.6):

$$\psi'' + (\varepsilon - \xi^2)\psi = e^{-\xi^2/2} [y'' - 2\xi y' + (\xi^2 - 1)y + (\varepsilon - \xi^2)y] = e^{-\xi^2/2} [y'' - 2\xi y' + (\varepsilon - 1)y].$$

The Gaussian prefactor never vanishes, so

$$y'' - 2\xi y' + (\varepsilon - 1)y = 0. \tag{6.4.7}$$

A power series  $y = \sum a_j \xi^j$  has coefficient recurrence

$$a_{j+2} = \frac{2j + 1 - \varepsilon}{(j+2)(j+1)} a_j.$$

# Quantum harmonic oscillator (cont.)

**Why choose a terminating series?** A bound state requires  $\int_{-\infty}^{\infty} |\psi|^2 d\xi < \infty$ . A polynomial  $y$  times  $e^{-\xi^2/2}$  satisfies this requirement. In contrast, a nonterminating parity chain has

$$\frac{a_{j+2}}{a_j} \sim \frac{2}{j},$$

like the coefficients of  $e^{\xi^2}$ . This suggests a growing exponential component that the prefactor  $e^{-\xi^2/2}$  cannot suppress. This coefficient comparison is intuition, not a full asymptotic proof. We use the standard bound-state result: decay at both ends selects a terminating chain, with the other parity chain absent.

Termination at degree  $n$  means the numerator  $2j + 1 - \varepsilon$  vanishes at  $j = n$ , i.e.

$$\varepsilon = 2n + 1, \quad n = 0, 1, 2, \dots,$$

and then  $\varepsilon - 1 = 2n$ , so (6.4.7) is exactly the Hermite equation  $y'' - 2\xi y' + 2ny = 0$ . Since  $\varepsilon = 2E/(\hbar\omega)$ , the energies are therefore

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, \dots, \quad (6.4.8)$$

and the spatial factors are  $\psi_n \propto e^{-\xi^2/2} H_n(\xi)$ .

# Generating function, norm, and Rodrigues

**Step 1: generating function from the series.** Define

$$g(x, t) := \exp(2xt - t^2) = e^{2xt} e^{-t^2}.$$

Expand each exponential:

$$e^{2xt} = \sum_{m=0}^{\infty} \frac{(2x)^m}{m!} t^m, \quad e^{-t^2} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} t^{2j}.$$

The product is a double sum. The coefficient of  $t^n$  arises from pairs  $(m, j)$  with  $m + 2j = n$ , i.e.  $m = n - 2j$  for  $j = 0, \dots, \lfloor n/2 \rfloor$ :

$$\begin{aligned} [t^n]g(x, t) &= \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(2x)^{n-2j}}{(n-2j)!} \cdot \frac{(-1)^j}{j!} \\ &= \frac{1}{n!} \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j (2x)^{n-2j} \frac{n!}{j! (n-2j)!}. \end{aligned}$$

# Generating function, norm, and Rodrigues (cont.)

Comparing with the series solution (6.4.3), the sum in brackets is exactly  $H_n(x)$ .  
Therefore

$$g(x, t) = e^{-t^2+2xt} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}. \quad (6.4.9)$$

**Step 2: Rodrigues from  $g = e^{x^2} e^{-(x-t)^2}$ .** Rewrite the exponent:

$$2xt - t^2 = x^2 - (x - t)^2, \quad g(x, t) = e^{x^2} e^{-(x-t)^2}.$$

Taylor-expand in  $t$  about  $t = 0$ :

$$e^{-(x-t)^2} = \sum_{n=0}^{\infty} \frac{t^n}{n!} \left. \frac{\partial^n}{\partial t^n} e^{-(x-t)^2} \right|_{t=0}.$$

On the composite  $e^{-(x-t)^2}$ , one has  $\partial/\partial t = -\partial/\partial x$ , so

$$\left. \frac{\partial^n}{\partial t^n} e^{-(x-t)^2} \right|_{t=0} = (-1)^n \frac{d^n}{dx^n} e^{-x^2}.$$

# Generating function, norm, and Rodrigues (cont.)

Hence

$$g(x, t) = e^{x^2} \sum_{n=0}^{\infty} \frac{t^n}{n!} (-1)^n \frac{d^n}{dx^n} e^{-x^2} = \sum_{n=0}^{\infty} \left[ (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \right] \frac{t^n}{n!}.$$

Matching coefficients with (6.4.9) gives Rodrigues' formula

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}. \quad (6.4.10)$$

**Step 3: recurrences from  $\partial_t g$  and  $\partial_x g$ .** Differentiate the closed form:

$$\frac{\partial g}{\partial t} = (2x - 2t)g, \quad \frac{\partial g}{\partial x} = 2tg.$$

Using the series (6.4.9),

$$\partial_t g = \sum_{n=1}^{\infty} H_n \frac{t^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} H_{n+1} \frac{t^n}{n!}.$$

Multiplication by  $t$  shifts the index in the other term:

$$2tg = \sum_{n=1}^{\infty} 2H_{n-1} \frac{t^n}{(n-1)!} = \sum_{n=1}^{\infty} 2nH_{n-1} \frac{t^n}{n!},$$

# Generating function, norm, and Rodrigues (cont.)

where  $1/(n-1)! = n/n!$ . Thus the coefficient of  $t^n/n!$  in  $(2x - 2t)g$  is  $2xH_n - 2nH_{n-1}$ . Comparing with  $\partial_t g$  gives, for  $n \geq 1$ ,

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x). \quad (6.4.11)$$

Likewise,  $\partial_x g = 2tg$ . The same power shift just used gives

$$\sum_{n=0}^{\infty} H'_n(x) \frac{t^n}{n!} = \sum_{n=1}^{\infty} 2nH_{n-1}(x) \frac{t^n}{n!}.$$

Comparing coefficients of  $t^n/n!$  gives

$$H'_n(x) = 2nH_{n-1}(x), \quad n \geq 1. \quad (6.4.12)$$

The constant coefficient separately gives  $H'_0 = 0$ .

# Generating function, norm, and Rodrigues (cont.)

**Step 4: norm via the double generating function.** For real  $s, t$  near 0,

$$I(s, t) := \int_{-\infty}^{\infty} e^{-x^2} g(x, s) g(x, t) dx = \int_{-\infty}^{\infty} e^{-x^2 - s^2 + 2sx - t^2 + 2tx} dx.$$

Complete the square in  $x$ :

$$-x^2 + 2(s+t)x - s^2 - t^2 = -[x - (s+t)]^2 + 2st,$$

so

$$I(s, t) = \sqrt{\pi} e^{2st} = \sqrt{\pi} \sum_{k=0}^{\infty} \frac{(2st)^k}{k!} = \sqrt{\pi} \sum_{k=0}^{\infty} \frac{2^k}{k!} s^k t^k.$$

On the other hand, expand both generating functions:

$$g(x, s) g(x, t) = \sum_{m,n=0}^{\infty} H_m(x) H_n(x) \frac{s^m t^n}{m! n!},$$

For  $s, t$  near zero, the Gaussian decay permits termwise integration:

$$I(s, t) = \sum_{m,n=0}^{\infty} \left( \int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx \right) \frac{s^m t^n}{m! n!}.$$

# Generating function, norm, and Rodrigues (cont.)

The right-hand closed form  $\sqrt{\pi} e^{2st}$  contains only diagonal powers  $s^k t^k$ . Equating coefficients of  $s^m t^n$ :

- if  $m \neq n$ , the right side has coefficient 0, so  $\int H_m H_n e^{-x^2} dx = 0$ ;
- if  $m = n$ , compare  $[s^n t^n]$ :

$$\frac{1}{(n!)^2} \int_{-\infty}^{\infty} H_n^2 e^{-x^2} dx = \sqrt{\pi} \frac{2^n}{n!},$$

Multiply by  $(n!)^2$  to obtain the diagonal norm.

Combining both cases,

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = 2^n n! \sqrt{\pi} \delta_{mn}. \quad (6.4.13)$$

Low- $n$  check:  $H_0 = 1$  and  $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ .

# Useful formulas for the Hermite polynomial

differential equation  $H_n'' - 2x H_n' + 2n H_n = 0$

series solution 
$$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k (2x)^{n-2k} \frac{n!}{k! (n-2k)!}$$

generating function 
$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$$

recurrence  $H_n' = 2n H_{n-1}, \quad H_{n+1} = 2x H_n - 2n H_{n-1}$

orthogonality 
$$\int_{-\infty}^{\infty} H_m H_n e^{-x^2} dx = n! 2^n \sqrt{\pi} \delta_{mn}$$

Rodrigues 
$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$$

Here  $n \geq 0$ ; the displayed recurrences use  $n \geq 1$ , with  $H_0 = 1$ ,  $H_1 = 2x$ .

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# Laguerre equation, polynomials, and orthogonality

Standard Laguerre polynomials solve

$$x y'' + (1 - x)y' + n y = 0, \quad n = 0, 1, 2, \dots \quad (6.5.1)$$

Generalized Laguerre polynomials occur in the radial hydrogen-atom equation. Here we study only the standard family, defined by (6.5.1) and normalized by  $L_n(0) = 1$ .

**Step 1: power series and coefficient recurrence.** Dividing by  $x$  gives  $P = (1 - x)/x$  and  $Q = n/x$ , so  $\lim_{x \rightarrow 0} xP = 1$  and  $\lim_{x \rightarrow 0} x^2Q = 0$ . The Chapter 3 indicial test gives  $r(r - 1) + r = r^2 = 0$ . The finite branch is therefore an ordinary power series; the second solution has a logarithmic singularity. We keep the branch regular at the origin. The polynomials grow at infinity, but the weight  $e^{-x}$  makes their squared norms finite.

# Laguerre equation, polynomials, and orthogonality (cont.)

Set  $y = \sum_{k=0}^{\infty} a_k x^k$ . Then

$$y' = \sum_{k=0}^{\infty} (k+1) a_{k+1} x^k,$$

$$xy'' = \sum_{k=0}^{\infty} k(k+1) a_{k+1} x^k$$

$$\left( \text{from } y'' = \sum (k+2)(k+1) a_{k+2} x^k, \text{ times } x, \text{ reindex} \right),$$

$$-xy' = -\sum_{k=0}^{\infty} k a_k x^k,$$

$$ny = \sum_{k=0}^{\infty} n a_k x^k.$$

Collecting the coefficient of  $x^k$  in  $xy'' + (1-x)y' + ny$  gives

$$k(k+1)a_{k+1} + (k+1)a_{k+1} + (n-k)a_k = (k+1)^2 a_{k+1} + (n-k)a_k.$$

# Laguerre equation, polynomials, and orthogonality (cont.)

Setting every coefficient to zero yields the one-step recurrence

$$a_{k+1} = -\frac{n-k}{(k+1)^2} a_k, \quad k = 0, 1, 2, \dots \quad (6.5.2)$$

At  $k = n$  one has  $a_{n+1} = 0$ , so the series terminates as a polynomial of degree  $n$ . With the conventional choice  $a_0 = 1$ , iterating (6.5.2) gives

$$a_k = (-1)^k \frac{n(n-1) \cdots (n-k+1)}{(k!)^2} = (-1)^k \frac{1}{k!} \binom{n}{k},$$

and therefore

$$L_n(x) = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!}. \quad (6.5.3)$$

The first few:

$$L_0 = 1, \quad L_1 = 1 - x, \quad L_2 = 1 - 2x + \frac{1}{2}x^2, \quad L_3 = 1 - 3x + \frac{3}{2}x^2 - \frac{1}{6}x^3.$$

**Step 2: Sturm–Liouville form and off-diagonal orthogonality.** Multiplying (6.5.1) by  $e^{-x}$  produces the SL form  $(xe^{-x}y')' + ne^{-x}y = 0$ . The same boundary-vanishing

# Laguerre equation, polynomials, and orthogonality (cont.)

argument as for Hermite (polynomials and their derivatives are finite at zero, and  $e^{-x}$  dominates their growth at infinity) gives, for  $n \neq m$ ,

$$\int_0^\infty L_n(x)L_m(x)e^{-x} dx = 0. \quad (6.5.4)$$

**Step 3: generating function and the diagonal norm.** Define

$$g(x, t) := \frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} L_n(x) t^n, \quad |t| < 1. \quad (6.5.5)$$

The first equality is the definition of  $g$ ; the second is a claim about the polynomials (6.5.3), and we prove it now. Expand the exponential and absorb the prefactor:

$$\frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \frac{1}{1-t} \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{-xt}{1-t}\right)^k = \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} \frac{t^k}{(1-t)^{k+1}}.$$

The binomial series from Chapter 1 gives

$$(1-t)^{-(k+1)} = \sum_{j=0}^{\infty} \binom{k+j}{k} t^j, \quad |t| < 1.$$

# Laguerre equation, polynomials, and orthogonality (cont.)

So the power  $t^n$  is produced by the pairs with  $j = n - k$  and  $0 \leq k \leq n$ , each carrying the factor  $\binom{k+(n-k)}{k} = \binom{n}{k}$ :

$$[t^n]g(x, t) = \sum_{k=0}^n \frac{(-x)^k}{k!} \binom{n}{k} = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!} = L_n(x)$$

by (6.5.3). The power series is absolutely convergent for  $|t| < 1$ , which permits the rearrangement.

Use the double-generating-function method just applied to Hermite. For real  $s, t$  near zero, put

$$\alpha = 1 + \frac{s}{1-s} + \frac{t}{1-t} = \frac{1-st}{(1-s)(1-t)} > 0.$$

Then

$$e^{-x}g(x, s)g(x, t) = \frac{e^{-\alpha x}}{(1-s)(1-t)}.$$

Since  $\int_0^\infty e^{-\alpha x} dx = 1/\alpha$ , the integral becomes

$$\int_0^\infty e^{-x}g(x, s)g(x, t) dx = \frac{1}{(1-s)(1-t)\alpha} = \frac{1}{1-st}.$$

# Laguerre equation, polynomials, and orthogonality (cont.)

Near  $s = t = 0$ , the exponential decay permits term-by-term integration. Expand both sides. The right side is

$$\frac{1}{1-st} = \sum_{n=0}^{\infty} (st)^n = \sum_{n=0}^{\infty} s^n t^n.$$

The left side is

$$\sum_{m,n=0}^{\infty} \left( \int_0^{\infty} L_m(x) L_n(x) e^{-x} dx \right) s^m t^n.$$

Equating coefficients of  $s^m t^n$ : if  $m \neq n$  the integral vanishes (already known); if  $m = n$  the coefficient is 1. Therefore the family is orthonormal:

$$\int_0^{\infty} L_m(x) L_n(x) e^{-x} dx = \delta_{mn}. \quad (6.5.6)$$

**Step 4: three-term recurrence from  $\partial_t g$ .** From the closed form (6.5.5),

$$\frac{\partial g}{\partial t} = g \cdot \frac{1-t-x}{(1-t)^2},$$

# Laguerre equation, polynomials, and orthogonality (cont.)

or equivalently

$$(1-t)^2 \partial_t g = (1-t-x) g. \quad (6.5.7)$$

Expand both sides in powers of  $t$ . With  $g = \sum_{n \geq 0} L_n t^n$  one has  $\partial_t g = \sum_{n \geq 0} (n+1) L_{n+1} t^n$ , so the left side of (6.5.7) is

$$(1-2t+t^2) \sum_{n \geq 0} (n+1) L_{n+1} t^n.$$

The coefficient of  $t^n$  on the left is

$$(n+1)L_{n+1} - 2nL_n + (n-1)L_{n-1}$$

(for  $n \geq 1$ , with the convention  $L_{-1} := 0$ ). On the right,  $(1-t-x) \sum L_n t^n$  contributes

$$(1-x)L_n - L_{n-1}.$$

Equating coefficients and rearranging yields

$$(n+1)L_{n+1}(x) = (2n+1-x)L_n(x) - nL_{n-1}(x). \quad (6.5.8)$$

# Laguerre equation, polynomials, and orthogonality (cont.)

Check at  $n = 1$ :  $(2)L_2 = (3 - x)L_1 - L_0$  gives  $2(1 - 2x + \frac{1}{2}x^2) = (3 - x)(1 - x) - 1$ , which holds identically.

A second useful identity is a derivative recurrence. Since

$$g_x = -\frac{t}{1-t}g,$$

the minus sign must be retained. A convenient coefficient comparison combines this identity with  $g_t$ :

$$\begin{aligned}\sum_{n \geq 1} n(L_n - L_{n-1})t^n &= t(1-t)g_t - tg \\ &= -\frac{xt}{1-t}g = xg_x = \sum_{n \geq 0} xL'_n(x)t^n.\end{aligned}$$

Therefore, for  $n \geq 1$ ,

$$xL'_n(x) = n[L_n(x) - L_{n-1}(x)]. \quad (6.5.9)$$

# Laguerre equation, polynomials, and orthogonality (cont.)

**Step 5: Schläfli integral and Rodrigues formula.** For any circle  $|z| = \rho < 1$ , Cauchy coefficient extraction applied to (6.5.5) gives a contour representation with the correct power of  $z$ :

$$L_n(x) = \frac{1}{2\pi i} \oint_{|z|=\rho} \frac{e^{-xz/(1-z)}}{(1-z)z^{n+1}} dz. \quad (6.5.10)$$

For  $x \neq 0$ , set  $z = (s - x)/s = 1 - x/s$ . Then

$$1 - z = \frac{x}{s}, \quad dz = \frac{x}{s^2} ds, \quad -\frac{xz}{1-z} = x - s,$$

and the image contour encloses  $s = x$  counterclockwise. Substitution gives

$$L_n(x) = \frac{e^x}{2\pi i} \oint \frac{s^n e^{-s}}{(s - x)^{n+1}} ds.$$

Cauchy's derivative formula now yields the Rodrigues construction

$$L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n} (x^n e^{-x}). \quad (6.5.11)$$

# Laguerre equation, polynomials, and orthogonality (cont.)

The identity at  $x = 0$  follows by continuity (and gives  $L_n(0) = 1$ ). For  $n = 2$ , two product-rule steps give

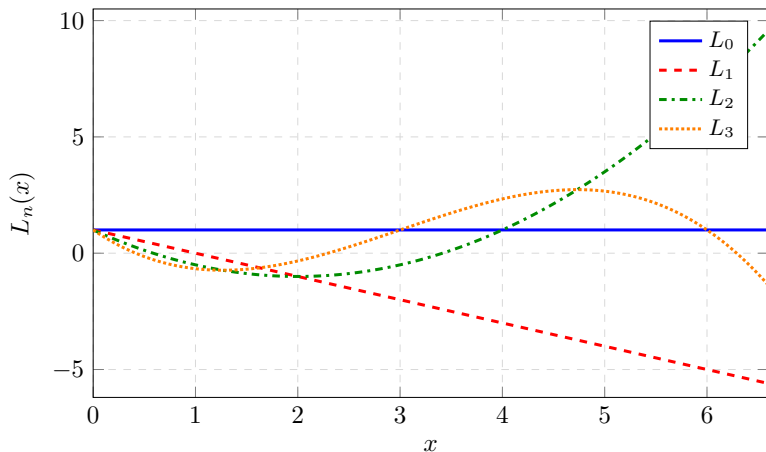
$$\begin{aligned}\frac{d}{dx}(x^2 e^{-x}) &= (2x - x^2)e^{-x}, \\ \frac{d^2}{dx^2}(x^2 e^{-x}) &= (2 - 4x + x^2)e^{-x}, \\ L_2(x) &= \frac{1}{2}(2 - 4x + x^2) = 1 - 2x + \frac{x^2}{2}.\end{aligned}$$

For general  $n$ , the product rule distributes  $k$  derivatives to  $e^{-x}$  and  $n - k$  to  $x^n$ , giving

$$\frac{e^x}{n!} \sum_{k=0}^n \binom{n}{k} \frac{n!}{k!} x^k (-1)^k e^{-x} = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!} = L_n(x).$$

Thus the derivative formula reproduces exactly the series already obtained.

# Shapes of the first Laguerre polynomials



All standard Laguerre polynomials satisfy  $L_n(0) = 1$ . The weight  $e^{-x}$  makes them natural on  $[0, \infty)$ .

# Useful formulas for the Laguerre polynomial

differential equation  $xL_n'' + (1-x)L_n' + nL_n = 0$

series / first polynomials 
$$L_n(x) = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!}$$

$$L_0 = 1, L_1 = 1 - x, L_2 = 1 - 2x + \frac{1}{2}x^2$$

generating function 
$$\frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} L_n(x) t^n$$

recurrence 
$$(n+1)L_{n+1} = (2n+1-x)L_n - nL_{n-1}$$

$$xL_n' = n(L_n - L_{n-1})$$

orthonormality 
$$\int_0^{\infty} L_m L_n e^{-x} dx = \delta_{mn}$$

Rodrigues 
$$L_n = \frac{e^x}{n!} \frac{d^n}{dx^n} (x^n e^{-x})$$

Generating series requires  $|t| < 1$ .

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# Definition through $\cos n\theta$

The Chebyshev polynomial of the first kind is defined by the elementary identity

$$T_n(\cos \theta) = \cos(n\theta). \quad (6.6.1)$$

That is, set  $x = \cos \theta$  and read off a polynomial in  $x$ :

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_2(x) = 2x^2 - 1, \quad T_3(x) = 4x^3 - 3x.$$

(For  $T_2$ :  $\cos 2\theta = 2\cos^2 \theta - 1$ . For  $T_3$ :  $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ .)

**Recurrence from angle addition.** Adding  $\cos((n+1)\theta) = \cos(n\theta)\cos\theta - \sin(n\theta)\sin\theta$  and  $\cos((n-1)\theta) = \cos(n\theta)\cos\theta + \sin(n\theta)\sin\theta$  cancels the sines:

$$T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x).$$

Thus

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \quad n \geq 1, \quad (6.6.2)$$

with  $T_0 = 1$ ,  $T_1 = x$ . Each step raises the degree by one: the leading term of  $2xT_n$  cannot cancel with  $T_{n-1}$ . Starting with leading coefficient 1 for  $T_1$ , it doubles at every step, so  $T_n$  has leading coefficient  $2^{n-1}$  for  $n \geq 1$ .

## Definition through $\cos n\theta$ (cont.)

The endpoints follow by setting  $\theta = 0, \pi$ :  $T_n(1) = 1$  and  $T_n(-1) = (-1)^n$ . Replacing  $\theta$  by  $\pi - \theta$  changes  $x$  to  $-x$ , so

$$T_n(-x) = \cos(n\pi - n\theta) = (-1)^n T_n(x).$$

**Generating function from the recurrence.** A single power series packages all the  $T_n$  at once. Put

$$G(x, t) := \sum_{n=0}^{\infty} T_n(x) t^n.$$

For  $x \in [-1, 1]$ ,  $|T_n(x)| \leq 1$ , so comparison with  $\sum |t|^n$  allows regrouping when  $|t| < 1$ . Multiply (6.6.2) by  $t^{n+1}$  and sum over  $n \geq 1$ , which is the full range in which the recurrence holds:

$$\sum_{n=1}^{\infty} T_{n+1} t^{n+1} = 2xt \sum_{n=1}^{\infty} T_n t^n - t^2 \sum_{n=1}^{\infty} T_{n-1} t^{n-1}.$$

## Definition through $\cos n\theta$ (cont.)

Each sum is  $G$  with a few terms removed or added. Using  $T_0 = 1$  and  $T_1 = x$ ,

$$\begin{aligned}\sum_{n=1}^{\infty} T_{n+1} t^{n+1} &= \sum_{k=2}^{\infty} T_k t^k = G - 1 - xt, & \sum_{n=1}^{\infty} T_n t^n &= G - 1, \\ \sum_{n=1}^{\infty} T_{n-1} t^{n-1} &= \sum_{k=0}^{\infty} T_k t^k = G.\end{aligned}$$

So  $G - 1 - xt = 2xt(G - 1) - t^2 G$ ; collecting the  $G$  terms on the left and the rest on the right gives  $(1 - 2xt + t^2)G = 1 + xt - 2xt = 1 - xt$ , that is

$$\sum_{n=0}^{\infty} T_n(x) t^n = \frac{1 - xt}{1 - 2xt + t^2}, \quad |t| < 1, \quad -1 \leq x \leq 1. \quad (6.6.3)$$

**Differential equation via the chain rule.** Set  $x = \cos \theta$  and  $Y(\theta) := T_n(\cos \theta) = \cos(n\theta)$ , so  $Y''(\theta) = -n^2 \cos(n\theta) = -n^2 Y(\theta)$ . Chain rule with  $dx/d\theta = -\sin \theta$  gives

$$\frac{d}{d\theta} = -\sin \theta \frac{d}{dx}, \quad \frac{dY}{d\theta} = -\sin \theta T'_n(x).$$

## Definition through $\cos n\theta$ (cont.)

Differentiate again:

$$\begin{aligned}\frac{d^2 Y}{d\theta^2} &= \frac{d}{d\theta} (-\sin \theta T'_n(x)) \\ &= -\cos \theta T'_n(x) - \sin \theta \cdot (-\sin \theta T''_n(x)) \\ &= -\cos \theta T'_n(x) + \sin^2 \theta T''_n(x).\end{aligned}$$

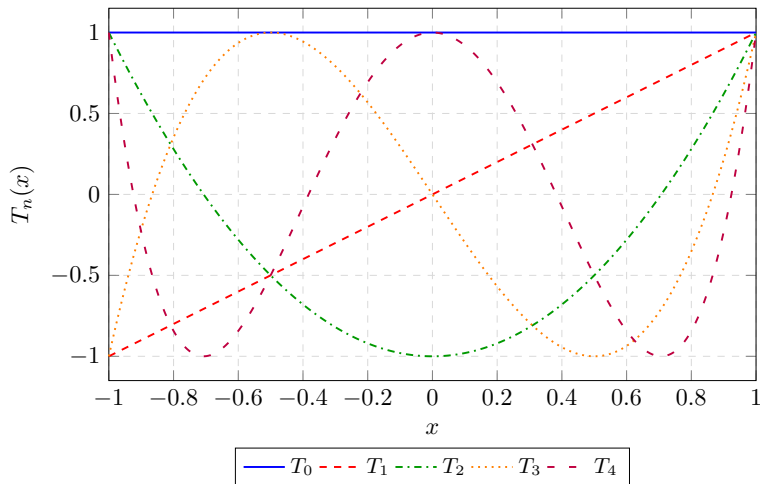
Substitute  $\cos \theta = x$  and  $\sin^2 \theta = 1 - x^2$ , and use  $Y'' = -n^2 Y = -n^2 T_n$ :

$$(1 - x^2) T''_n(x) - x T'_n(x) = -n^2 T_n(x).$$

Rearranging yields the Chebyshev equation

$$(1 - x^2) T''_n(x) - x T'_n(x) + n^2 T_n(x) = 0. \quad (6.6.4)$$

# Shapes of the first Chebyshev polynomials



$T_n(\cos \theta) = \cos(n\theta)$  keeps every  $T_n$  between  $-1$  and  $1$  on  $[-1, 1]$ .

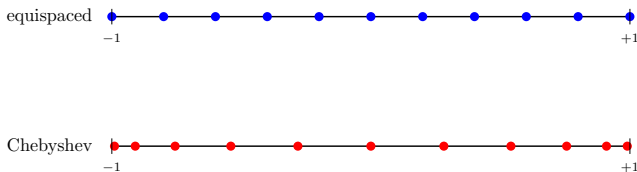
# Chebyshev nodes and orthogonality

**Zeros (nodes).** For  $n \geq 1$  and  $x = \cos \theta \in [-1, 1]$ ,

$$T_n(x) = 0 \iff \cos(n\theta) = 0 \iff \theta_k = \frac{(2k-1)\pi}{2n}, \quad k = 1, \dots, n.$$

$$x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \quad k = 1, \dots, n. \quad (6.6.5)$$

The angles are equally spaced, but  $x = \cos \theta$  crowds the nodes near  $\pm 1$ . This endpoint clustering is useful for polynomial interpolation.



# Chebyshev nodes and orthogonality (cont.)

**Orthogonality.** Dividing the differential equation by  $\sqrt{1-x^2}$  gives

$$\left(\sqrt{1-x^2} T_n'\right)' + \frac{n^2}{\sqrt{1-x^2}} T_n = 0.$$

Thus the natural weight is  $w(x) = 1/\sqrt{1-x^2}$  (up to a constant factor). More intuitively, this weight cancels the change-of-variable factor in  $x = \cos \theta$ , turning the integral into ordinary cosine orthogonality.

With the weight  $(1-x^2)^{-1/2}$ , substitute  $x = \cos \theta$ . Then  $dx = -\sin \theta d\theta$ , and as  $x$  runs from  $-1$  to  $1$ ,  $\theta$  runs from  $\pi$  down to  $0$ . Also  $\sqrt{1-x^2} = \sqrt{1-\cos^2 \theta} = \sin \theta$  for  $\theta \in [0, \pi]$ . Therefore

$$\begin{aligned} \int_{-1}^1 \frac{T_n(x) T_m(x)}{\sqrt{1-x^2}} dx &= \int_{\pi}^0 \frac{T_n(\cos \theta) T_m(\cos \theta)}{\sin \theta} (-\sin \theta) d\theta \\ &= \int_0^{\pi} T_n(\cos \theta) T_m(\cos \theta) d\theta \\ &= \int_0^{\pi} \cos(n\theta) \cos(m\theta) d\theta, \end{aligned}$$

## Chebyshev nodes and orthogonality (cont.)

where the last step is the defining identity  $T_k(\cos \theta) = \cos(k\theta)$ . For  $n \neq m$ , use  $\cos A \cos B = [\cos(A - B) + \cos(A + B)]/2$ :

$$\int_0^\pi \cos(n\theta) \cos(m\theta) d\theta = \frac{1}{2} \left[ \frac{\sin((n-m)\theta)}{n-m} + \frac{\sin((n+m)\theta)}{n+m} \right]_0^\pi = 0.$$

For  $n = m \geq 1$ , use  $\cos^2(n\theta) = [1 + \cos(2n\theta)]/2$ :

$$\int_0^\pi \cos^2(n\theta) d\theta = \left[ \frac{\theta}{2} + \frac{\sin(2n\theta)}{4n} \right]_0^\pi = \frac{\pi}{2}.$$

For  $n = m = 0$ , the integrand is 1 and the integral is  $\pi$ . Hence

$$\int_{-1}^1 \frac{T_n(x) T_m(x)}{\sqrt{1-x^2}} dx = \begin{cases} \pi, & n = m = 0, \\ \pi/2, & n = m \geq 1, \\ 0, & n \neq m. \end{cases} \quad (6.6.6)$$

# Useful formulas for the Chebyshev polynomial

|                       |                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| definition            | $T_n(\cos \theta) = \cos(n\theta)$                                                                                                      |
| differential equation | $(1 - x^2)T_n'' - xT_n' + n^2T_n = 0$                                                                                                   |
| first polynomials     | $T_0 = 1, T_1 = x, T_2 = 2x^2 - 1, T_3 = 4x^3 - 3x$                                                                                     |
| recurrence            | $T_{n+1} = 2xT_n - T_{n-1}$                                                                                                             |
| generating function   | $\frac{1 - xt}{1 - 2xt + t^2} = \sum_{n=0}^{\infty} T_n(x)t^n, \quad (6.6.3)$                                                           |
| zeros                 | $x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right)$                                                                                           |
| orthogonality         | $\int_{-1}^1 \frac{T_n T_m}{\sqrt{1-x^2}} dx = \begin{cases} \pi, & n = m = 0, \\ \pi/2, & n = m \geq 1, \\ 0, & n \neq m. \end{cases}$ |

Generating series requires  $|t| < 1$ .

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## Summary: four orthogonal families

| Family          | Interval            | Weight             | ODE (sketch)                          |
|-----------------|---------------------|--------------------|---------------------------------------|
| Legendre $P_n$  | $[-1, 1]$           | 1                  | $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$ |
| Hermite $H_n$   | $(-\infty, \infty)$ | $e^{-x^2}$         | $y'' - 2xy' + 2ny = 0$                |
| Laguerre $L_n$  | $[0, \infty)$       | $e^{-x}$           | $xy'' + (1 - x)y' + ny = 0$           |
| Chebyshev $T_n$ | $[-1, 1]$           | $(1 - x^2)^{-1/2}$ | $(1 - x^2)y'' - xy' + n^2y = 0$       |

**Physical origins (one line each).**  $P_n$ : axisymmetric Laplace / multipoles;  $H_n$ : quantum harmonic oscillator;  $L_n$ : its generalized family occurs in the hydrogen radial equation;  $T_n$ :  $\cos n\theta$  and stable interpolation.

The shared structure is a second-order Sturm–Liouville ODE, a recurrence, and an orthogonality relation on a natural interval and weight. Orthogonality alone only tells us how to *compute* the coefficients of a series; to know that the series actually reproduces  $f$  we need the separate property of *completeness*, stated for  $P_n$  on the Fourier–Legendre slide. It holds for all four families, as stated on the next slide, and then lets us expand reasonable functions exactly as Fourier series expand in  $e^{inx}$ .

# Why the expansions converge: completeness

Orthogonality finds the coefficients. Completeness says that the full family can approximate every function with a finite weighted squared norm.

For any of the four families, let  $\phi_n$  denote its polynomials and use the interval  $I$  and weight  $w$  from the table. Then

$$c_n = \frac{\int_I f(x) \phi_n(x) w(x) dx}{\int_I \phi_n(x)^2 w(x) dx}, \quad S_N(x) = \sum_{n=0}^N c_n \phi_n(x).$$

The completeness result, used here without proof, is

$$\int_I |f(x) - S_N(x)|^2 w(x) dx \longrightarrow 0 \quad \text{whenever} \quad \int_I |f(x)|^2 w(x) dx < \infty.$$

This is *mean-square* convergence: the total weighted squared error tends to zero. It does not assert convergence at every individual point.

**The common workflow.** Choose the family from the differential equation and its interval; identify the weight; compute coefficients by orthogonality; use a finite sum as an approximation.